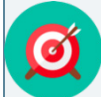


11.4 Graphing Exponential Functions Given a Written Description and Equation

Lesson Introduction

 Benchmarks	MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay. MA.912.AR.5.6 Given a table, equation, or written description of an exponential function, graph that function and determine its key features.		 Learning Targets	- I can graph an exponential function given a written description and equation. - I can determine the key features of an exponential function. - I can classify a function as representing exponential growth or decay.
 Benchmark Coherence	Building On		Working Toward	
	N/A		MA.912.AR.5.7: Solve and graph mathematical and real-world problems that are modeled with exponential functions. Interpret key features and determine constraints in terms of the context.	
 Essential Question	How can you graph an exponential function given a written description and equation?			
 Lesson Narrative	In this lesson, students extend what they learned about graphing exponential functions from equations and tables of values to graphing a given written description. Students will be introduced to how to graph an exponential function from a written description and equation and then collaborate on the process. They will then transition to a gallery walk, where they will apply this knowledge to given real-world situations.			
 Component Summary	11.4.1 Warm-Up (~5 min.)	Would you rather? activity (SE p. 176)	11.4.3 Collaboration (10 min.)	Collaboration activity on graphing exponential functions given a written description and equation (SE p. 178)
	11.4.2 Guided Instruction (10 min.)	Introduction to graphing exponential functions given a written description and equation (SE p. 176)	11.4.4 Activity (15-20 min.)	Gallery walk: graphing an exponential function given a written description and equation (SE p. 179)
	11.4.5 Wrap-Up (~5 min.)	Growth mindset self-reflection (SE p. 180)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		- Poster board - Graph paper (Optional) Chart paper (Optional) Markers	The gallery walk in 11.4.4 Activity will require materials such as posterboard, graph paper, chart paper, and markers. Consider predetermining your number of groups and having “kits” with the needed material ready to avoid loss of instructional time.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle to interpret key features from a written description. - Students may mistake a decay situation with a growth situation given written descriptions. - Students may struggle to determine what the constant percent of change is from a written description. - Students may misinterpret the y-intercept in context.	- This lesson extends what students learned about graphing exponential functions to real-world contexts. Consider leveraging word problem and literacy strategies to assist students with understanding the information that is being provided in each problem. - Work through the activities to determine pacing that makes the most sense based on your students' level of understanding at this point in the unit. For example, 11.4.3 Collaboration could be assigned for homework if you think students are ready to work on this without guidance, which would allow more time to incorporate the gallery walk activity, if desired.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads This lesson consists primarily of word problems and contexts that require the students to fully understand what is occurring in the problem. Have students engage in this protocol to ensure students are aware of the full context prior to attempting to solve the given problem.	MA.K12.MTR.7.1: Real-World Contexts This lesson consists of a collection of problems where exponential functions need to be graphed based on a written description and equation that are based in a real-world context. Students will use models and methods to understand, represent, and solve problems throughout this lesson.
 Differentiate Instruction	Since this lesson requires a greater degree of reading comprehension, consider grouping your students for 11.4.4 Activity by their reading level to work with them in smaller groups at the stages in the process where they may need your assistance the most.	
	For 11.4.4 Activity, provide students with poster board or other display methods where students can hang their work up around the room for all to see as part of the gallery walk process.	
Student performance during 11.4.4 Activity can provide valuable insight into students' levels of understanding of the content of this lesson. This can come from the scenario they create the graph for or the feedback and conversation they have at the other posters.		
Lesson Notes		

11.4.1 Warm-Up: Would You Rather?

Component Overview: Students will engage in a “Would you rather?” scenario to determine whether they would choose the option of receiving a one-time payment of \$1 million dollars or receiving an increasing number of pennies for one month and justify their answer mathematically.



11.4.1 Warm-Up: Would You Rather?

- Imagine you have the chance to pick from one of the two options shown.

Option A	Option B
Receive a one-time payment of \$1 million dollars. 	Receive an increasing number of pennies each day as shown. OR  on day 1 on day 2 on day 3 and so on for 1 month

- Would you rather choose Option A or Option B?

Answers will vary.

- Using mathematics, show which option pays out more money.

$$0.01(2)^{30} = \$10,737,418.24$$



This is generally an engaging discussion to have with students. Some students will catch on quickly and realize that the answer is probably Option b, but they may not be able to mathematically justify why in question 1b. Consider creating an opportunity for whole-group discussion of these options to elicit student responses from multiple different perspectives.

11.4.2 Guided Instruction: Graphing Exponential Functions From a Written Description and Equation

Component Overview: Students will be introduced to graphing exponential functions from a written description and equation. They will be guided step by step through an example real-world problem.

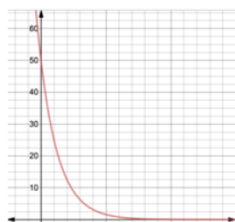


11.4.2 Guided Instruction: Graphing Exponential Functions from a Written Description and Equation

The half-life of a drug is an estimated period of time that it takes for an amount of that drug in the body to be reduced by exactly half. The amount, A , in milligrams, of a drug in the body for h hours after taking a pill is given by

$$A(h) = 50 \left(\frac{1}{2}\right)^h.$$

1. Graph the function that models the situation.



2. Describe how to determine whether the situation models exponential growth or decay.

The graph declines at a rapid rate and approaches the x -axis, which shows it is decay.

3. What is the initial dose given?

50 milligrams

4. Describe how to identify the initial dose from the function and from the graph in the table.

From the Function	From the Graph
<i>From the a value</i>	<i>From the y-intercept</i>



Students may be unfamiliar with the term “half-life.” Have a brief discussion with students to ensure that all students are familiar with the meaning of the term.



- *How can you determine the initial dose given?*
- *Where in the equation can you locate the initial dose given?*
- *Where on the graph can you locate the initial dose given?*

11.4.2 Guided Instruction: Graphing Exponential Functions From a Written Description and Equation (continued)

5. What percentage of the drug leaves the body each hour?
50%
6. How is this percentage represented in the equation?
As the b value, $\frac{1}{2}$
7. Complete the table to identify other key features of the function based on the context. In the second row, explain how each key feature relates to the context.

	Domain	Range	As x -values increase, y -values ...
Key Feature	$h \geq 0$	$0 < A(h) \leq 50$	<i>Decrease</i>
Reason for Constraint from Context	<i>The number of hours can't be negative.</i>	<i>The amount of a drug in someone's body cannot be negative and will be less than or equal to 50 milligrams.</i>	<i>A medicine in the body would not remain in the body forever.</i>



How do you determine what percentage of the drug leaves the body each hour?



Provide an opportunity for students to share out and discuss the constraints part of this table. It's important that constraints consistently be explained and discussed when they arise, as it is an area that is often overlooked by students in pursuit of the correct answer.

11.4.3 Collaboration: Graphing Exponential Functions

Component Overview: Students will work with a partner to graph an exponential function given a written description and equation.



11.4.3 Collaboration: Graphing Exponential Functions

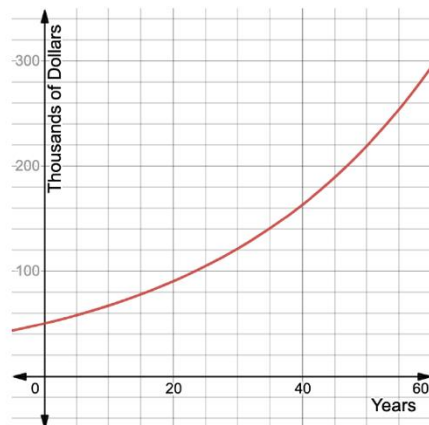
Liam recently graduated from the University of South Florida and accepted a job as a software engineer. The starting salary is \$50,000 and has a guaranteed annual pay increase of 3% each year.

The equation $y = 50,000(1 + 0.03)^x$ models the increase in salary over time, where x is the number of years since he started at the company and y is his salary.

- Describe how to determine if this situation models exponential growth or decay.

It is an exponential growth because the value of b is greater than 1.

- Graph the equation $y = 50,000(1 + 0.03)^x$.



- What does the y -intercept represent in this situation?
Liam's starting salary.



When partnering students up for this Collaboration activity, it does not have to be the same groupings you want to use for the upcoming card sort activity. Consider watching and listening to students during the previous 11.4.2 Guided Instruction activity to determine which students may need additional support with this concept.

11.4.3 Collaboration: Graphing Exponential Functions (continued)

4. Harper said that the domain is $0 \leq x \leq \infty$. Discuss with your partner if Harper is correct. Write a summary of your discussion.

Harper is correct because the number of years he is employed with the company cannot be less than 0.

5. Describe why the range depends on the number of years the person works for the company.

Liam receives a raise every year, therefore the number of years will determine how much he will make after he receives his annual increase.

6. Complete the statements.

If Liam works for the company for 30 years, the domain is

$0 \leq x \leq 30$, with a range of $50,000 \leq y \leq 121,363.12$.

7. Two students were asked to find the constant percent rate of change for this situation. Their answers are shown.

Owen	Fiona
The constant percent rate of change for the situation is 3% because Liam receives a pay increase of 3% every year.	The constant percent rate of change is 103% because each year Liam receives a 3% pay increase in addition to 100% of the salary he earned the year before.

- a. Discuss with your partner who identified the correct constant percent rate.

Fiona is correct because his increase signifies a growth of 3%, meaning he keeps his 100% and grows an addition 3%.

- b. Explain the mistake made by the student who was incorrect.

Owen saw the 3% as the change, not realizing that would represent a 97% loss when calculating.



Questions 4 and 5 make for great mathematical discussions as a whole group. Consider asking students to share their thinking on these questions and leveraging an agree/disagree protocol to have students continue the discussion a bit more.



- What would be a scenario in which Owen would be correct?
- If someone said that Fiona can't be correct because a percent only goes to 100, how would you respond?

11.4.4 Activity: Gallery Walk

Component Overview: Students will work in groups to graph a situation that has been provided for them. Each group will post their work in different locations in the room and move from poster to poster, viewing others' work and providing feedback.



11.4.4 Activity: Gallery Walk

Work with a partner to graph the situation below that your teacher has chosen for you.

Situation A	Situation B
The annual rate of inflation for goods and services is modeled by $C(t) = P(1.024)^t$, where C is the approximate cost of goods or services, t is the time in years, and P is the present cost. If the present cost of Nike LeBron basketball shoes is \$149.99, graph the function that will model the cost of shoes over time.	A new 2021 Ford Ranger XLT costs \$31,045 with a depreciation rate of 8% per year. This situation is modeled by the function $V(x) = 31,045(1 - 0.08)^x$, where x is the number of years since 2021 and V is the value of the car.
Situation C	Situation D
One hundred bacteria cells are put into a petri dish, and the culture doubles in size every hour. This is modeled by $P(h) = 100(2)^h$, where P is the number of bacteria in the population and h is the number hours that have passed.	A new computer that was recently purchased depreciates over time by the given function, $C(x) = \$1,200\left(\frac{1}{4}\right)^x$, where x is the number of years since the computer was purchased and C is the current value of the computer.



For this activity, students should graph their situation on a poster board, chart paper, or graph paper, which should be hung up around the room or passed around from group to group.

Below each graph, students should include the key features and their interpretations of their problem. Do not give students the key features; see what they come up with on their own.

Once each group has completed their graph, they will move around the room, commenting on things the other groups did well and any corrections (including missing key features) to other groups' work.

If movement is an issue, consider having each group present their situation and elicit feedback from their peers.

11.4.4 Activity: Gallery Walk (continued)

Once you and your partner graph the function, determine the key features and interpret how each relates to the context.

Situation A.

	Key Feature	Interpretations
Domain	$0 \leq t < \infty$	<i>The domain is the time at the initial sale and goes until the shoes are no longer sold.</i>
Range	$149.99 \leq C(t) < \infty$	<i>The range starts at the initial price and goes until the shoes are no longer sold.</i>
y-intercept	$(0, 149.99)$	<i>This represents the price at the initial sale.</i>
Constant Percent Rate of Change	$C(t) = 149.99(1.024)^t$ $1.024 \times 100 = 102.4\%$	<i>This represents the rate of inflation over time.</i>
End Behavior	$t \rightarrow \infty, C \rightarrow \infty$	<i>As time continues, the price goes up with inflation.</i>



Students may need additional assistance with understanding what is to be placed in the "Interpretations" column.



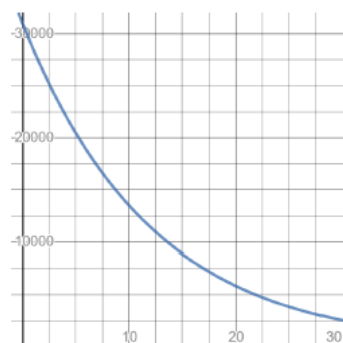
11.4.4 Activity: Gallery Walk (continued)

Situation B.

Key Feature		Interpretations
Domain	$0 \leq x < \infty$	The domain starts in 2021 and goes until the truck is no longer sold.
Range	$0 < V(x) \leq 31045$	The range starts at the initial price and goes until the truck no longer has any value.
y-intercept	(0,31045)	This represents the price at the initial sale.
Constant Percent Rate of Change	$V(x) = 31045(1 - 0.08)^x$ $V(x) = 31045(0.92)^x$ $0.92 \times 100 = 92\%$	This represents the rate of depreciation over time.
End Behavior	$x \rightarrow \infty, V \rightarrow 0$	As time continues, the price goes down with depreciation.



Consider printing the answer keys for each situation out so that you have them handy as you circulate the room to monitor and assist students during this process.



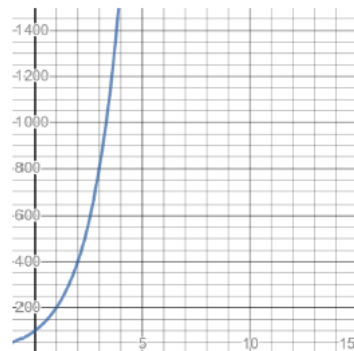
11.4.4 Activity: Gallery Walk (continued)

Situation C.

Key Feature		Interpretations
Domain	$0 \leq h < \infty$	<i>The domain starts when the initial bacteria are put in the petri dish and increases each hour.</i>
Range	$100 \leq P < \infty$	<i>The range starts at the initial bacteria count and continues to increase until the experiment concludes.</i>
y-intercept	(0,100)	<i>This represents the initial amount of bacteria put in the petri dish.</i>
Constant Percent Rate of Change	$P(h) = 100(2)^h$ $2 \times 100 = 200\%$	<i>This represents the rate of increase in the number of bacteria.</i>
End Behavior	$h \rightarrow \infty, P \rightarrow \infty$	<i>As time continues, the amount of bacteria will increase until stopped.</i>



While circulating, consider asking probing questions when students are getting off track. If students miss a key feature, try not to tell them what they are missing. Instead, challenge them with a question to consider. Part of the beauty of this activity will be the peer feedback.



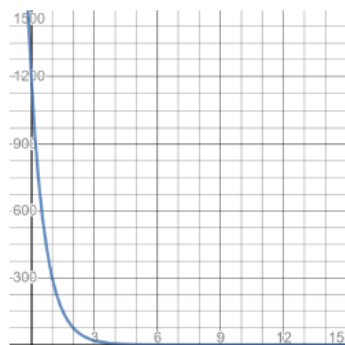
11.4.4 Activity: Gallery Walk (continued)

Situation D.

Key Feature		Interpretations
Domain	$0 \leq x < \infty$	The domain starts with the purchase of the computer and increases over time.
Range	$0 < C \leq 1200$	The domain is the initial price of the computer and decreases as the value depreciates.
y-intercept	(0,1200)	This represents the initial price of the computer.
Constant Percent Rate of Change	$C(x) = \$1,200 \left(\frac{1}{4}\right)^x$ $\frac{1}{4} \times 100 = 25\%$	This represents the rate of decrease in the value of the computer.
End Behavior	$x \rightarrow \infty, C \rightarrow 0$	As time continues, the value of the computer decreases until it no longer holds value.



If a group finishes early, consider having them verify their graph using a form of graphing technology.



11.4.5 Wrap Up: Growth Mindset

Component Overview: Students will complete a self-reflection related to growth mindset during this lesson.



11.4.5 Wrap-Up: Growth Mindset

Answers will vary.

Which mindset did I demonstrate?	Choose the best description.
Did I use the partner discussions as learning opportunities?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I work on problems that challenged me?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I put as much effort as I possibly could into the problems?	☐ Never ☐ Sometimes ☐ Usually ☐ Always



What would reflection look like for you as a teacher? Consider your own informal data regarding students' understanding of the concepts taught so far in this unit and how to best to address potential gaps in understanding prior to the unit assessment.

Consider making notes on the unit guide for things you would want to keep in mind for this unit next school year.